

APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Predicting first-stage recovery success from launch telemetry, using data collected via public API and web scraping, then modeled with classical ML classifiers.

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GitHub Repository



All notebooks, Python scripts, and SQL files for this project are available at:

<https://github.com/jai1116-lang/spacex-falcon9-capstone/tree/main>

- Data collection notebooks (API + web scraping)
- Data wrangling & labeling scripts
- EDA (SQL + visualization) notebooks
- Interactive visual analytics (Folium, Plotly Dash)
- Predictive analysis / classification notebook

Executive Summary

SpaceX advertises Falcon 9 launches at ~\$62M, versus \$165M+ for competitors — largely because SpaceX reuses the first stage. If we can predict whether a landing will succeed before launch, we can estimate launch cost and better evaluate a competing bid.

90

Launches analyzed

Falcon 9 only, Falcon 1 excluded

66.7%

Historical success rate

First-stage recovery, all sites

83.3%

Best model test accuracy

Decision Tree classifier

3

Launch sites

CCAFS, KSC, VAFB

Key finding

Landing success correlates strongly with orbit type and launch recency: later flights (higher flight number), lighter payloads, and orbits like VLEO/LEO land more reliably than heavy GTO missions.

Project Background & Questions

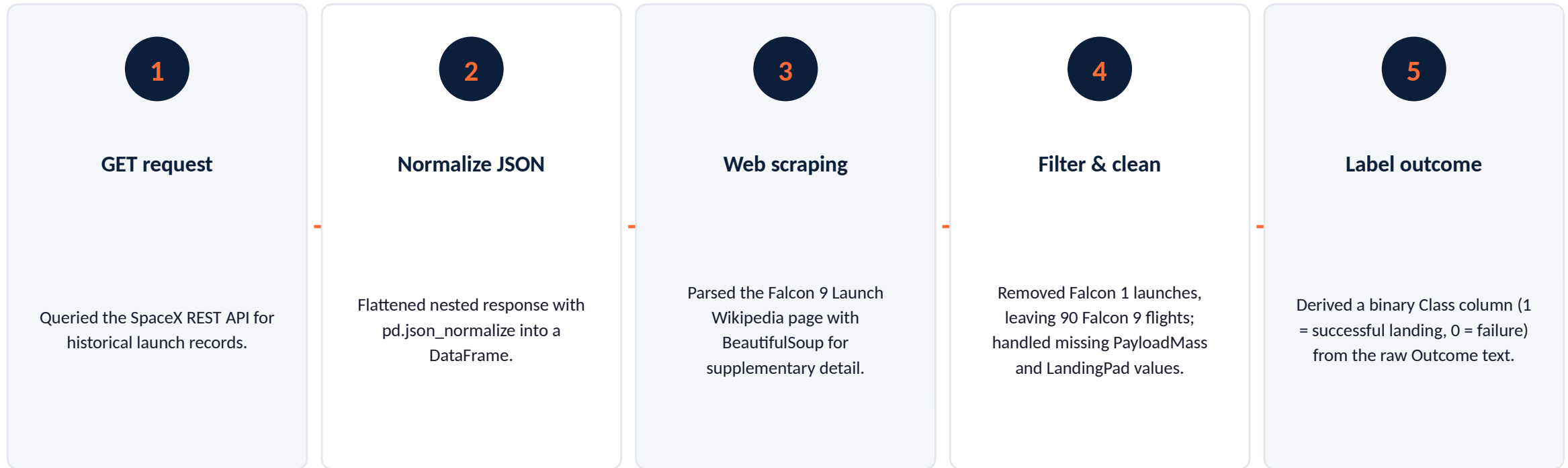
Background

SpaceX's reusable Falcon 9 first stage is the key driver of its low launch cost. Whether a first stage lands successfully depends on many factors — payload mass, orbit, launch site, booster generation, and more. This project builds a data pipeline and classification model to predict landing outcome from pre-launch data.

Questions this project answers

- 1 What features most influence landing success?
- 2 Does landing success improve over time / mission number?
- 3 Which launch sites and orbits have the best track record?
- 4 Can we predict landing outcome with a classifier — and how accurate is it?

Data Collection and Data Wrangling Methodology



Result: a clean, labeled dataset of 90 Falcon 9 launches with 17 columns spanning payload, orbit, site, booster, and landing outcome.

EDA and Data Visualization Methodology



Visual EDA (Seaborn / Matplotlib)

Scatter and bar plots: FlightNumber vs. PayloadMass, LaunchSite success rate, Orbit vs. Outcome, and yearly success-rate trend — used to spot which features correlate with landing success before modeling.



SQL EDA (Db2 / SQLite)

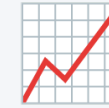
Loaded the cleaned launch data into a SQL table and queried launch-site counts, payload aggregates, orbit breakdowns, and outcome counts — cross-checking the pandas-based EDA with independent SQL results.

Interactive Visual Analytics Methodology



Folium Map

Plotted all 3 launch sites and every individual launch outcome (success/fail markers) on an interactive Leaflet-based map, with distance lines to the nearest coastline, railway, and city to assess siting constraints.



Plotly Dash Dashboard

Built an interactive Dash app with a launch-site dropdown and a payload-mass range slider, driving a live pie chart (success share by site) and scatter plot (payload vs. outcome).

Predictive Analysis (Classification) Methodology

1

Standardize features (StandardScaler) after one-hot encoding categoricals (Orbit, LaunchSite, LandingPad, Serial, GridFins, Reused, Legs) — 83 total columns.

2

Split into train/test sets (80/20), yielding 18 test samples.

3

Grid-search four classifiers — Logistic Regression, SVM, Decision Tree, KNN — with 10-fold cross-validation on the training set.

4

Select best hyperparameters per model by validation accuracy, then evaluate all four on the held-out test set.

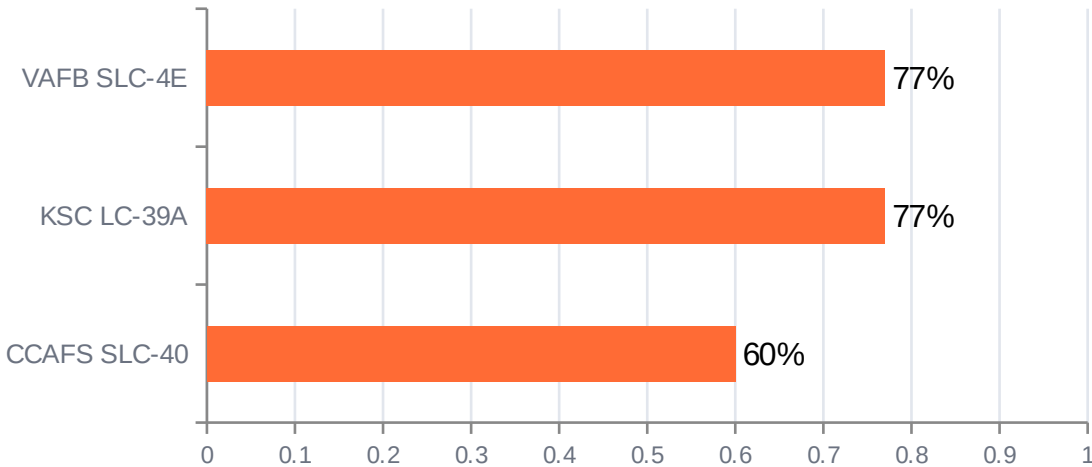
5

Compare test accuracy and confusion matrices to choose the final model.

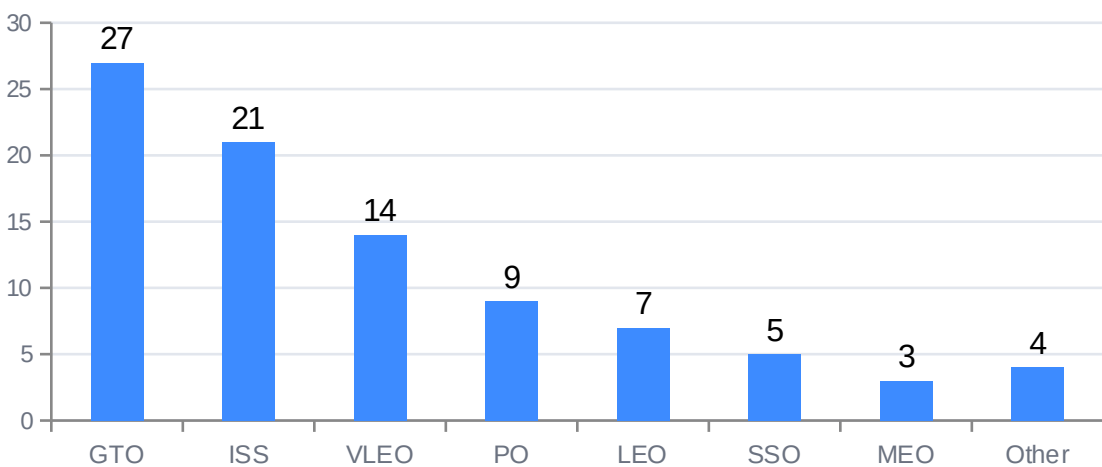
RESULTS

EDA with Visualization

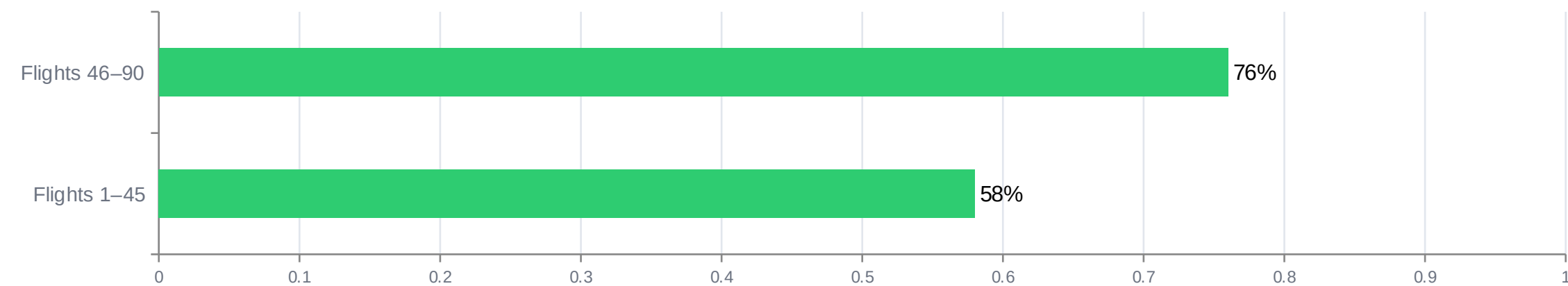
Success rate by launch site



Launch count by orbit type



Landing success improves with mission experience



RESULTS

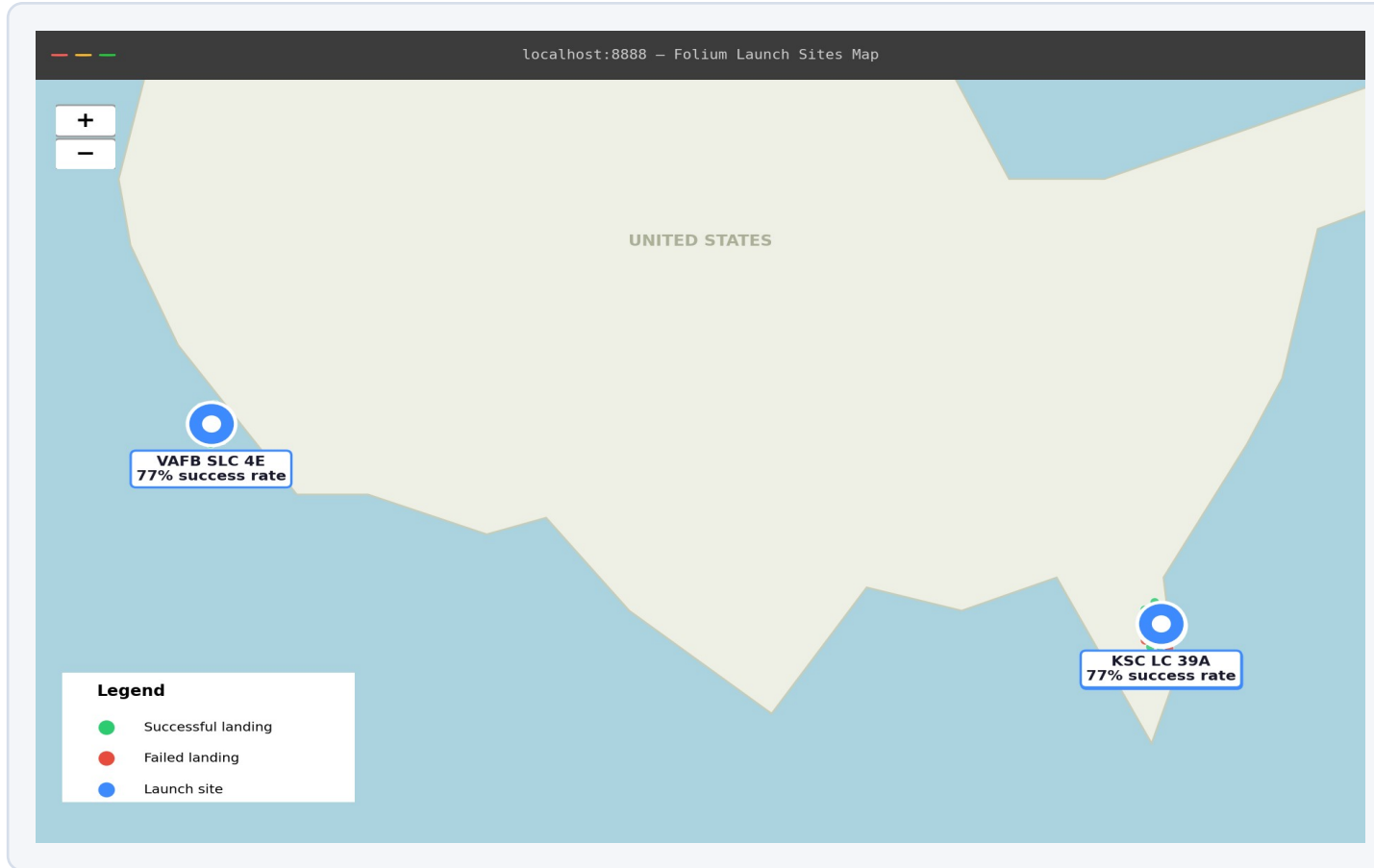
EDA with SQL Queries

SQL Query	Result
<pre>SELECT COUNT(*) FROM SPACEXTBL WHERE LaunchSite = 'CCAFS SLC 40'</pre>	55 launches
<pre>SELECT SUM(Class)*1.0/COUNT(*) FROM SPACEXTBL</pre>	66.7% success (60 of 90)
<pre>SELECT COUNT(*) FROM SPACEXTBL WHERE Orbit = 'GT0'</pre>	27 launches
<pre>SELECT COUNT(*) FROM SPACEXTBL WHERE Outcome = 'True ASDS'</pre>	41 drone-ship landings
<pre>SELECT DISTINCT LaunchSite FROM SPACEXTBL</pre>	CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
<pre>SELECT MIN(PayloadMass) FROM SPACEXTBL</pre>	350 kg

Full query set (9 queries) is in `sql/eda_queries.sql` and `notebooks/03_eda_sql.ipynb` in the GitHub repo.

RESULTS

Folium Map — Interactive Results

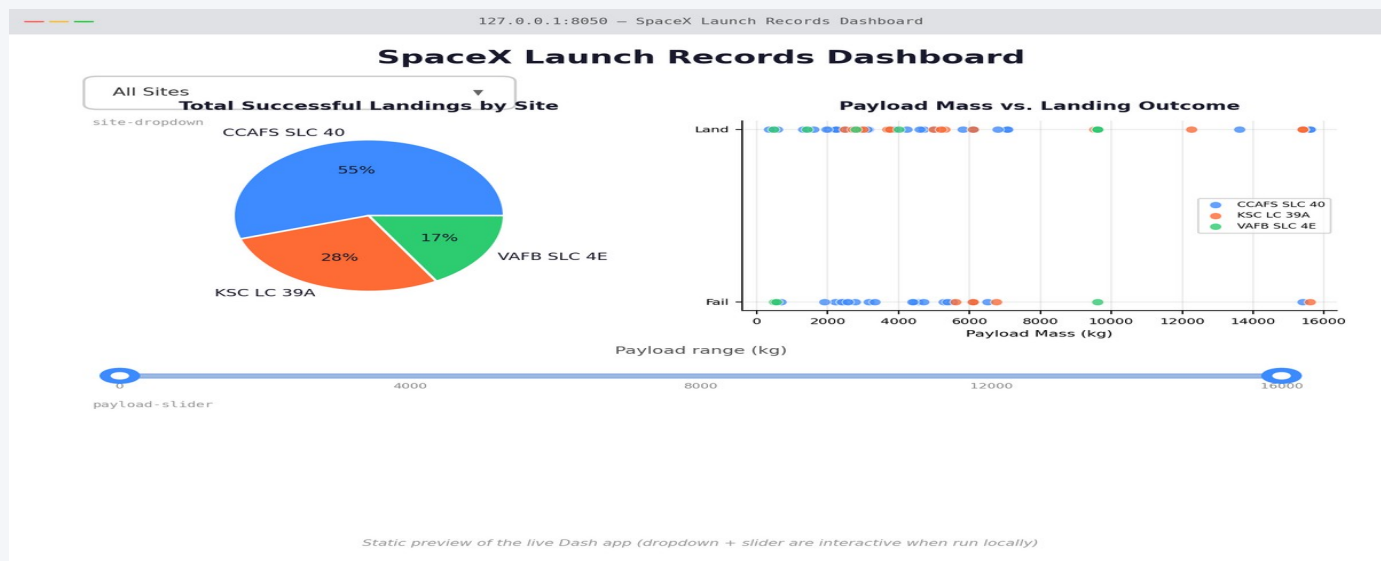


What the map shows

- Markers for all 3 launch sites, colored by aggregate success rate.
- Individual launch markers (green = success, red = failure) clustered per site.
- Distance lines from each site to the nearest coastline, railway, and city — used to assess siting constraints.
- Confirms launch sites are built near the coast, away from population centers, consistent with safety practice.

RESULTS

Plotly Dash — Interactive Dashboard



Static preview — the live app (notebook 5) adds an interactive site dropdown and payload-mass range slider that filter both charts.

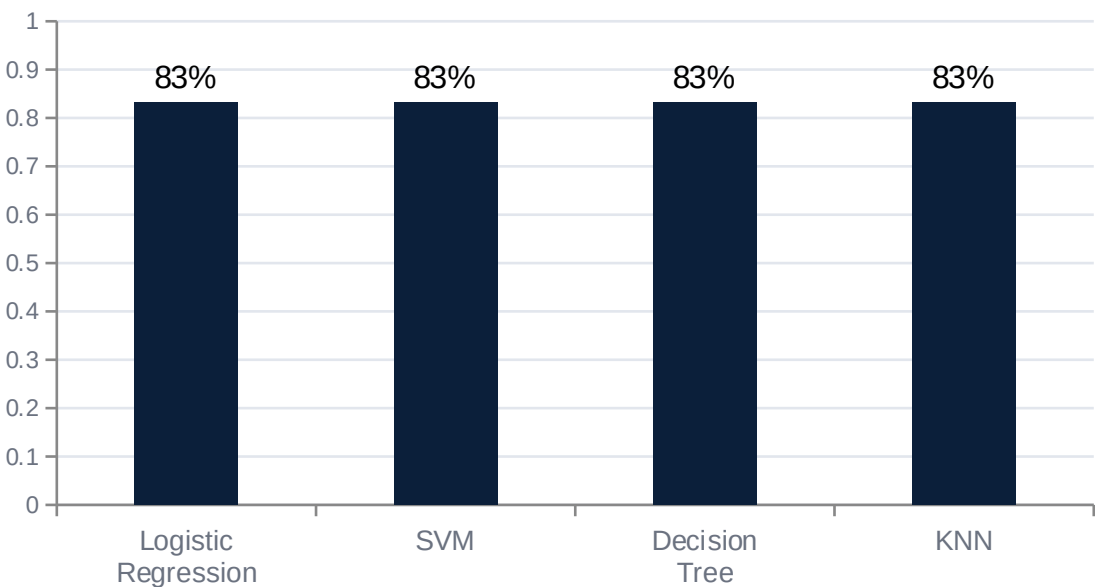
Dashboard features

- Dropdown to filter by launch site or view all sites combined.
- Pie chart of successful landings by site — KSC LC-39A leads at ~77%.
- Range slider to filter by payload mass (kg), updating the scatter plot live.
- Scatter plot of payload mass vs. landing outcome, colored by booster version.

RESULTS

Predictive Analysis Results

Test-set accuracy by model



All four models converge near the same test accuracy on this small (18-sample) held-out set — differences of 1–2 predictions shift the ranking, so cross-validation accuracy (below) is the more stable comparison.

Best model — Decision Tree

Metric	Value
Test accuracy	83.3%
Best criterion	gini
Best max_depth	6
Test samples	18 (20% holdout)

Confusion matrix (Decision Tree, test set)

	Pred: Fail	Pred: Land
Actual: Fail	5	1
Actual: Land	2	10

CONCLUSION

Key Takeaways

1

Landing success is learnable. A Decision Tree classifier reached 83.3% test accuracy predicting first-stage landing outcome from pre-launch features.

2

Experience matters. Success rate rose from ~58% in the first 45 launches to ~76% in the most recent 45 — SpaceX's landing process improved measurably over time.

3

Orbit and site matter. KSC LC-39A and VAFB SLC-4E outperform CCAFS SLC-40 (~77% vs. 60%); heavy GTO missions land less reliably than lighter LEO/VLEO missions.

4

Practical value. This kind of model lets a buyer estimate whether a quoted Falcon 9 launch will reuse its booster — and therefore estimate the likely price — before the mission flies.

Creativity & innovative insights: reframing landing prediction as a learning-curve story (58%→76% success as the mission count grows), not just a classifier score, plus a from-scratch dark-themed design and cross-validated SQL/visualization/ML results.

Next steps: incorporate weather data, add booster-generation features, and test ensemble methods for a further accuracy gain.